Clever

Lead Pipeline Reliability: Product Diagnosis & Execution Plan

Prepared by Hunter Doster · Diagnosing silent lead slips & sequencing the fix

01 Product Vision

Clever's lead pipeline is experiencing **silent failures**. Leads arrive from partner sources, enter the system, and disappear, with no alert, no record, and no one called. Engineering systems show green and cannot locate why leads are slipping through.

I propose the **Lead Slip Detector**, the product that solves this issue by first detecting all leads that slip, making sure the problem is no longer invisible, then creates the proper output and process to route these leads so they get contacted. The prototype is a Flask web app with a 6-tab dashboard that ingests lead and outreach data, identifies all slips, and surfaces them for immediate action. It is live on Railway and was deployed against the sample data provided.

Product Lifecycle

1. **Deploy Minimum Viable Product** →
2. **Define & Document Long-Term Solution** →
3. **Build & Deploy Long-Term Solution** →
4. **Detection & Maintenance**

The priority is speed. Deploy the MVP. Make the problem visible. Then build it right.

02 The Problem at a Glance

206	379	18	12	30	14.6%	20	345
LEADS REVIEWED	OUTREACH ATTEMPTS	NEVER CONTACTED (T1)	SMS FAIL, NO CALL (T1)	TOTAL TIER 1 SLIPS	NO-CONTACT RATE	DUPLICATE LEADS	SYNTAX ISSUES

Nearly 1 in 7 leads was never successfully contacted.

- Across **206 leads** over 13 days, **30 (14.6%)** were never successfully contacted — nearly 1 in 7.
- Those 30 break down as **18 never touched** (a silent OfferNest integration blackout) plus **12 failed-SMS leads** (landline) with no follow-up call.
- A further **20 leads** were duplicate contacts (the same customer reached twice).

Slip Cost — Leads Never Contacted (see Section 05 for methodology)

Revenue at Risk (3% conv.)	Per Month	Per Year
25% referral fee	$3,271	$39,799
40% referral fee	$5,234	$63,678
Estimated annual risk		**$40,000 – $64,000**

Answering the Three Core Questions

1 · How big? 14.6% of leads, nearly 1 in 7, were never successfully contacted. Of 206 leads over 13 days, 30 resulted in zero customer contact (18 never touched, 12 failed SMS with no follow-up). An additional 20 leads (9.7%) were duplicate contacts.

2 · Why? Three root causes: (1) a silent OfferNest failure created a 5-day blackout (May 11–16) dropping 18 leads; (2) 12 leads hit a landline SMS failure with no call escalation; (3) 10 duplicate pairs arrived cross-source with no resolution at intake.

3 · What's it costing? Using industry benchmarks (3% conversion, 1.5% fee, 30% referral, $420K median price), annualized revenue at risk from Tier 1 slips alone is **$40K–$64K/yr**. This estimate excludes duplicate-contact trust loss, partner attrition, and complaint cost, so the true cost is likely higher.

03 Definitions & Assumptions

A **lead slip** occurs when a lead enters the system but fails to result in customer contact. Slips are organized into three priority tiers:

Tier 1 (Highest) — Never Reached

These slips all produce the same outcome: the customer was never contacted. Highest priority because they represent potential business that was never reached.

- **Never Touched**: lead received from partner source but no outreach attempt is initiated.
- **Skipped by Dialer**: system receives lead but actively bypasses initiation without attempting contact. *(Clarification needed: is this detected differently than "Never Touched," or the same failure state with a different root cause?)*
- **System Constraint Failure**: SMS fails on landline detection (err30006_landline); lead cannot be reached via that channel, with no follow-up call.

Tier 2 (Medium) — Duplicate Contact

At least one outreach attempt was made; the problem is lead duplication, not total silence.

- **Duplicate Contact**: customer contacted multiple times for the same lead, indicating a conflict resolution or routing failure.

Tier 3 (Lowest) — Customer Feedback

- **No Contact Made**: outreach initiated but customer reports never hearing from us; follow-up chain breaks down.

Deliberately left alone (for now): Tier 3 customer-feedback complaints (no-contact via inbound channels) are not actioned in Phase 1. The assumption is that fixing Tier 1 slips resolves the majority of these. A complaint intake & tracking process will be scoped in Month 1.

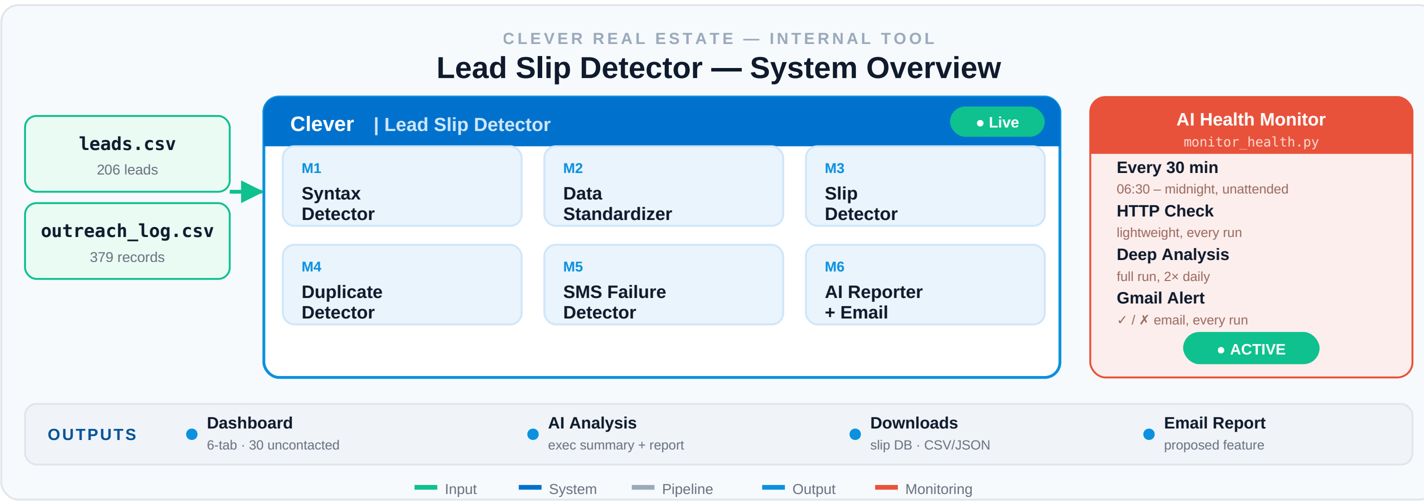
04 Analysis

Part 1 — Lead Slip Detector: Architecture & Modules LIVE ON RAILWAY

The six modules of the Lead Slip Detector run in sequence:

M1 · Syntax Detector	Flags data-formatting errors & structural inconsistencies in the CSVs that could block CRM integration.
M2 · Data Standardizer	Normalizes input into consistent JSON (phone formats, name casing, spacing), producing clean records for all downstream modules.
M3 · Slip Detector	One-to-one match of every lead against the outreach log; identifies leads with zero contact attempts (primary Type 1 detector).
M4 · Duplicate Detector	Finds matching contacts across partner sources, surfacing Tier 2 candidates.
M5 · SMS Failure Detector	Flags landline SMS failures (err30006) with no follow-up call. Classified Tier 1: attempt logged, customer never reached.
M6 · AI Reporter + Email	Aggregates slips through Claude Opus 4.8 (4 calls/run, ~15s): executive summary + 3-category pattern report; emails team on detection.

AI Health Monitoring (monitor_health.py): a dedicated monitor runs alongside the detector so it never goes dark undetected; health checks are emailed periodically. The self-contained static HTML demo exists at hunterdoster.com/clever-case-study-backend/, validated by 33 Playwright user-story tests.



Part 2 — Root Cause Analysis: Why Did This Happen?

RC1 — OfferNest Integration Failure (Type 1). All 18 never-touched leads came from a single source, **OfferNest**, with a clear five-day blackout beginning **May 11, 2026**. Likely causes (pending engineering review): API outage (leads queued/dropped silently), webhook drop (no error surfaced), routing misconfiguration (received but not queued), or dialer bypass. Confirming requires system logs from May 11–16. *However, root-cause analysis is not a prerequisite for the solution: the system should detect when any lead from any source is uncontacted past the slip threshold and alert automatically.*

RC2 — Landline SMS Failures, No Recovery (Type 2). 12 leads received an SMS that failed on landline detection (err30006_landline) with no follow-up call. There is no current mechanism to recognize an SMS failure and auto-escalate to a call.

RC3 — Duplicate Leads Across Partner Sources (Type 3). 10 lead group pairs (20 leads) where the same customer submitted through two partner sources. Every case was cross-source; no within-source duplicates. Phone normalization (last 10 digits) resolved all cleanly. Either customers submit through multiple channels, or partners share a lead database. Dual impact: the customer is contacted twice (trust degradation) and one copy may go unworked if routing treats it as already handled.

05 Cost Analysis

Methodology & Assumptions

Given the limited sample (206 leads / 13 days), industry benchmarks are used rather than extrapolating from this dataset alone. All figures will be replaced with Clever's actual internal metrics once available.

- **Median home price:** $420,000 (NAR Mar 2026, rounded up from $408,800, conservative forward estimate)
- **Clever listing fee:** 1.5% = $6,300/transaction (listwithclever.com)
- **Referral fee range:** 25%–40% of agent commission (HousingWire, Luxury Presence, The Close, 2026)
- **Lead-to-close conversion:** 3% (industry midpoint; range 1%–5%, Jamil Academy 2026)

Clever's actual conversion rate is unknown. 3% is a midpoint estimate and should be validated with internal data as a Week 1 priority.

How the Annual Figure Is Derived

The $40K–$64K range shown in Section 02 is built from the slip rate observed in the sample. From the sample (13 days): **18** leads with no outreach (OfferNest blackout) + **12** failed-SMS/no-call = **30 Tier 1 slips**. Annualized: ~2.3 slips/day → ~69/month → **~842/year**.

Applying a 3% conversion, the 1.5% listing fee ($6,300/transaction), and a 25%–40% referral fee to those ~842 annual slips yields the per-month and per-year revenue at risk. The range reflects the referral-fee band; the figure is conservative, excluding duplicate-contact trust loss, partner attrition, and complaint cost.

06 MVP Implementation Cost

The Lead Slip Detector is already live and operational. Below is the infrastructure cost to keep it running.

Component	Function	Plan	Monthly
Railway	Hosts Flask app (Dockerfile, gunicorn)	Hobby	$5
Claude Opus 4.8 API	AI slip analysis, 4 calls/run	PAYG	~$1–3
GitHub	Source control + auto-deploy	Free	$0
GitHub Pages	Static backup of the app	Free	$0
Gmail SMTP	Health-monitoring alerts	Free	$0
Total infrastructure			**~$6–8/mo**

Total Cost Summary

Cost Category	Annual Impact
Tier 1 slips, never contacted	$40K–$64K
Tier 2 slips, duplicate contact	TBD (trust risk)
Partner & customer complaints	TBD (attrition risk)
MVP infrastructure (annualized)	~$72–$96
Net: cost to fix vs. cost of inaction	**~$84/yr vs. $40K–$64K+**

Full cost picture to be completed in Week 1 once clarifying questions are answered.

Open Questions — Required to Refine Cost Picture and Long-Term Solution

- What is Clever's actual lead-to-close conversion rate?
- What is average revenue per closed lead (referral % and transaction value)?
- How many total leads does Clever receive per day / week / month? → needed for database sizing
- Does Clever track close rate by time-to-first-contact? (speed-to-lead impact)
- When duplicate contact occurs, how much does it degrade conversion?
- How many partner/customer complaints have been received about no-contact leads, and what was the impact?

07 Execution Plan

Slip Threshold Assumption. A lead is considered slipped if it has not been contacted within **60 minutes** of generation. In the sample data, median time-to-first-contact is 8 min and mean is 35.7 min, making 60 min a reasonable starting threshold.
To be confirmed with the team and revisited once broader operational data is available.

Week 1 — Deploy the MVP: Make Failures Visible

Goal: the problem is no longer invisible. Deploy the Lead Slip Detector as a Minimum Viable Process running automatically: pulling lead & outreach data, scanning for slips, alerting the team. An AI health-monitoring layer keeps the detector itself operational. Every slipped lead is caught and worked.

Current MVP output: the prototype already generates CSV + JSON of all detected slips per run, handed to sales for immediate follow-up until CRM integration exists.

Questions for Week 1:

- What CRM system is Clever using?
- How can slipped leads be pulled into the CRM?
- What is the process for routing a slipped lead back to the team?
- How are we notified of a lead-slip customer complaint? What is the current process and historical cost of this slip type?

End of Week 1: no lead slips without being caught. Detection is live and stress-tested; slips are documented, flagged, and actioned daily. The problem is no longer invisible.

Month 1 — Understand & Plan the Permanent Build

Goal: with the MVP operational and slips contained, begin diagnosis and planning for the permanent solution: understand internal infrastructure, document the full product, initiate the engineering build.

Questions for Month 1:

- What is Clever's current standard for first outreach?
- What contact speed is achievable given current staffing and dialer config?
- Should the threshold be 5 min (best practice), 60 min (current baseline), or in between?
- What infrastructure, dialer logic, and CRM config are required to hit that target?

End of Month 1: PRD finalized & signed off (incl. full build cost: money + resources); slip-contact-time standard agreed with leadership; engineering build initiated; slips still caught daily (MVP running).

Month 3 — Build & Deploy the Permanent System

Goal: full production deployment of a polished, industry-standard slip detection system. The MVP is retired. The system is fully automated, CRM-integrated, and running in real time: slips are detected and either fixed automatically or the team is notified immediately, with no manual process required.

Definition of Done — shipped at Month 3:

- Fully automated slip detection in real time, covering all Tier 1 & Tier 2 slips
- Slipped leads auto-pushed into CRM for immediate follow-up, with no manual CSV handoff
- Duplicate leads detected and handled in real time, with no customer contacted twice for the same inquiry
- Continuous monitoring dashboard live as a permanent operational resource
- **No slip is ever invisible again**

08 Engineering Requirements

Before accurate requirements can be defined, I need greater clarity on Clever's engineering team: structure, capability, tooling/languages, and how software gets built and shipped internally. That clarity comes in Month 1. Two options follow based on current knowledge; I believe Option 2 is the stronger path.

Option 1 · Traditional Software Development

Conventional team building & maintaining manually, without AI-assisted tooling.

Role	HC	Responsibility
Backend Eng	1–2	CRM integration, detection logic, data pipeline
Frontend Eng	1	Live monitoring dashboard & reporting UI
DevOps Eng	1	Deployment, scheduling, real-time infra, alerts
Total	**3–4**	**Full build: Month 2 – Month 3**

Option 2 · AI-Agent Driven Development PROPOSED

The industry is already moving here. OpenAI's Feb 2026 post "Harness engineering: leveraging Codex in an agent-first world" describes 3 engineers driving AI agents shipping ~1M lines / ~1,500 pull requests over five months, ~1/10th the time of hand-written code.

Role	Headcount
AI Agent Governing Engineer	1
Senior PM driving agent direction & requirements	1
Total	**2**

Why this project, why now? Annual risk from Tier 1 slips is **$40K–$64K**, a small, well-defined, fixable problem, which makes it the ideal first pilot for AI-agent driven development. Stakes are real but contained. The MVP is already built and the path from operational bridge to polished product is clear. The deployed MVP means our worst case while we build is fully contained: slips are caught and worked daily. Cost of doing nothing: $40K–$64K/yr. Cost of the MVP: minimal. Upside: a repeatable, scalable AI software development capability Clever can apply to every future product. Clever must adopt AI software-production methods to maintain growth and competitive edge; organizations that learn to build this way now will outpace traditional shops by an order of magnitude in speed, cost, and scale. This project is the right place to start.

09 Next Steps & Takeaways

This proposal asks leadership for two decisions:

Decision 1 — Engineering Approach

Which model builds the permanent system?

- **Option 1:** Traditional development (3–4 engineers, manual coding)
- **Option 2:** AI-agent driven development (1 engineer + 1 PM, agent-governed)

Decision 2 — Plan Approval

- **1. Approve & implement as-is**: greenlight MVP deployment now; proceed as written.
- **2. Changes needed, then implement**: specific adjustments to scope, sequencing, resourcing, or assumptions; then approved.
- **3. Larger changes required**: significant rework and a follow-up session before a path forward.